

Blockchain & The Digital Product Passport (DPP)

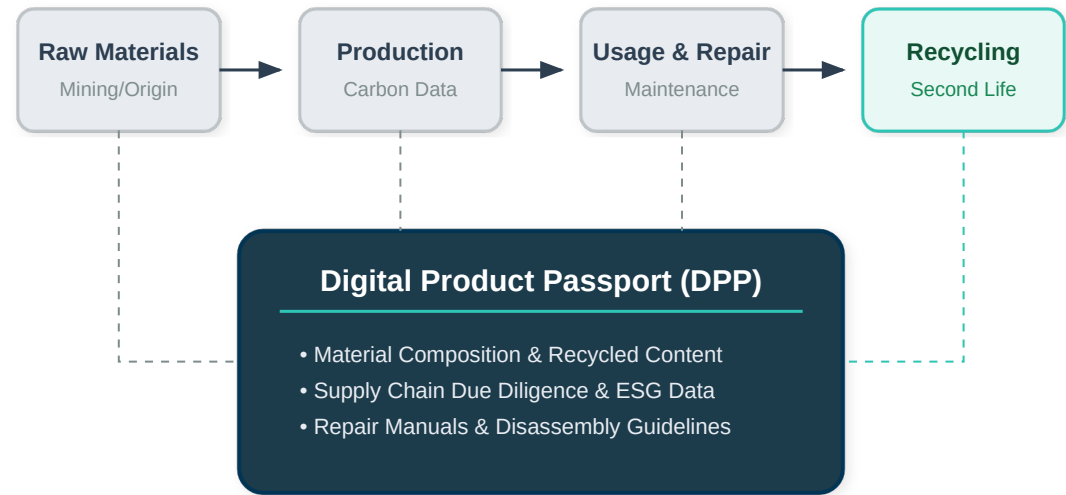
A Critical Examination of Technical Bottlenecks, Trust Assumptions, and Governance Silos

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The Digital Product Passport (DPP) Mandate

- **EU Ecodesign for Sustainable Products Regulation (ESPR)**
 - Requires a "Digital Twin" for batteries, electronics, textiles, and building materials
 - Mandates transparency on material composition, origin, recycling, and carbon footprint



The Promise vs. The Reality

The Promise ✨

- ✓ **Trustless Coordination**
Decentralized actors collaborate freely
- ✓ **Absolute Immutability**
No history can ever be falsified later
- ✓ **Seamless Product Transparency**
Full consumer insight from raw to recycled

The Reality ⚠️

- ✗ **Centralized Consortia**
Controlled by few industry oligarchs
- ✗ **Garbage In, Garbage Out**
Immutable storage of forged physical tags
- ✗ **IP & Trade Secret Leaks**
Exposes logistics, BOM, and supply chain partners

- **The Theory:**

- Standardized, frictionless, incorruptible decentralized ledger tracking circular lifecycle.

- **The Reality:**

- High costs, low performance, fragmentation into industry silos, privacy breaches, and zero

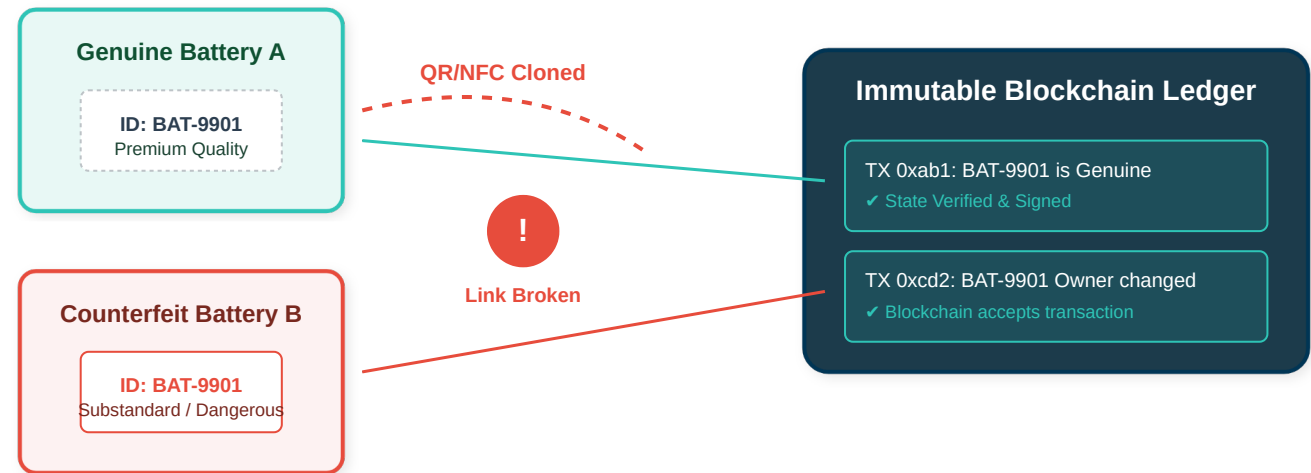
Issue #1: The Physical-Digital Linkage Problem

- The "Garbage In, Garbage Out" (GIGO) Dilemma
 - Ledgers are digital, products are physical.
 - Cryptographic signatures guarantee *who* wrote the data, not that the *physical reality* matches the entry.

- Vulnerabilities of Physical

Tags:

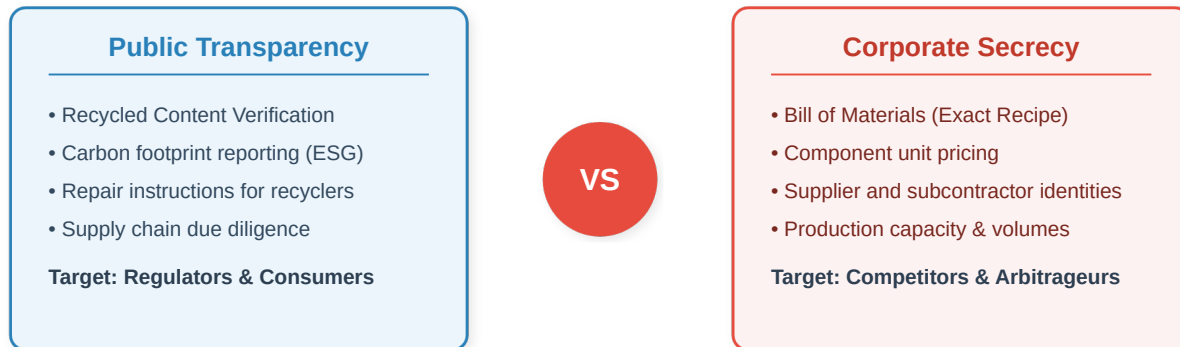
- QR Code Copying:



Issue #2: Privacy, Confidentiality, and Trade Secrets

The DPP Information Paradox

Auditable Transparency vs. Commercial Secrecy



- **The Transparency Paradox**
 - Regulators/consumers demand complete transparency of supply chains.
 - BUT: A product's **Bill of Materials (BOM)** and supplier registry are **critical trade secrets**.

- **Commercial Threats:**

Issue #3: Scalability, Latency, and Financial Costs

- **Transaction Volumes & Network Capacity**
 - EU markets process **billions** of new physical items each year.
 - Tracking every transit, repair, component swap, and recycling event demands high-throughput.
- **The Trilemma Constraint:**
 - **Public Blockchains (e.g., Ethereum L1/L2):** Fees are unpredictable and prohibitively high (e.g., \$0.05 to \$2.00 per update). Who pays to register a €3 garment's passport?
 - **Private Consortia:** High throughput and low fees, but they are just centralized databases managed by a cartel of large corporations.



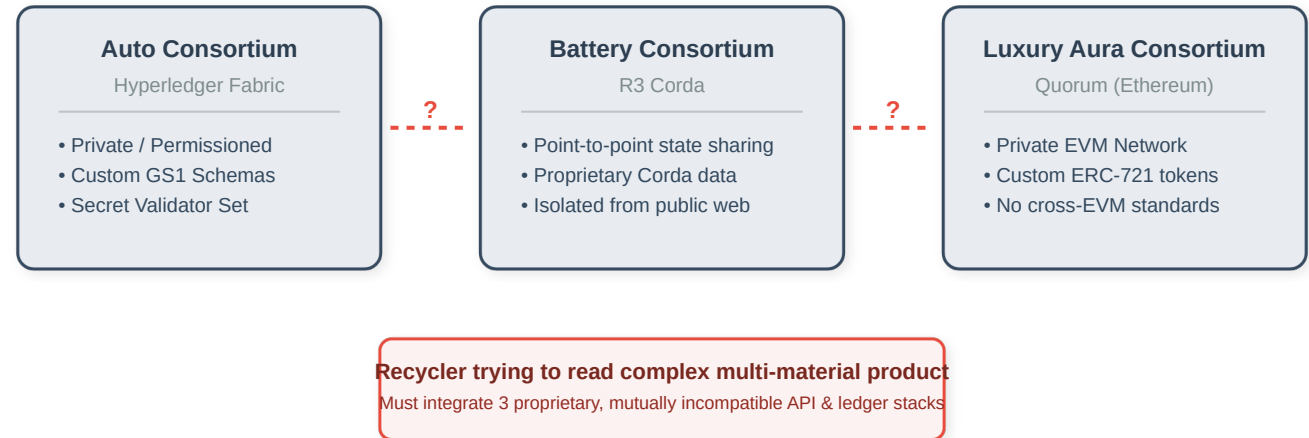
Issue #4: Governance, Standards, and Walled Silos

- **Fragmentation into Silos of Trust**

- There will never be a single, global "DPP Blockchain".
- Automotive (Catena-X), Luxury (Aura), and Tech (Battery Alliance) run separate, incompatible networks

The Fragmentation of Ledger Silos

Incompatible Networks & Divided Governance



Issue #5: GDPR vs. Blockchain Immutability

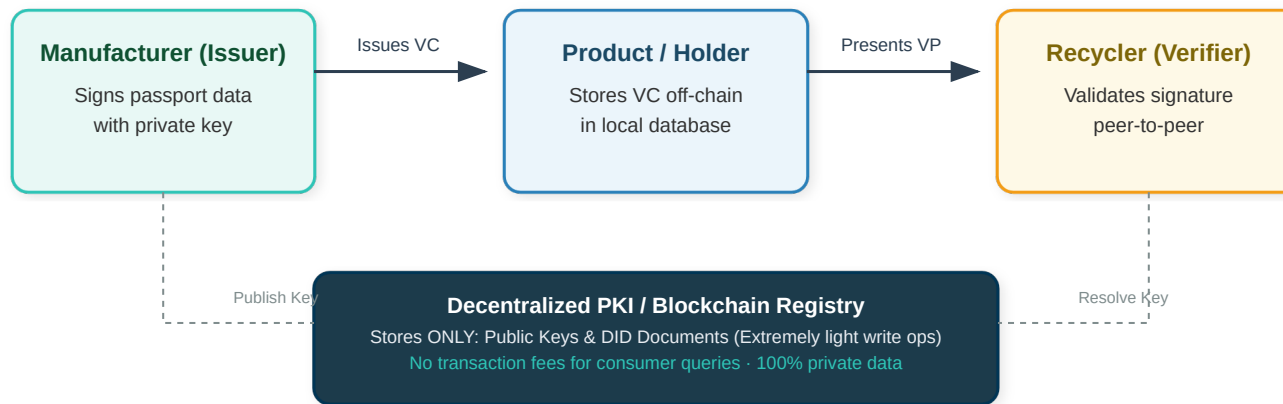
- **The Irreconcilable Clash of Law and Architecture**
 - **GDPR Article 17:** The "Right to be Forgotten" mandates the erasure of personal data on request.
 - **Blockchain Core Feature:** Permanent, non-deletable history.
- **How Personal Data Leaks Into the Passport:**
 - Ownership transfers of premium goods (e.g., e-bikes, luxury watches) identify buyers.
 - Repair and maintenance logs contain the names, licenses, and locations of technicians.

The Failure of "Off-Chain Hashing"

Alternative Architecture: Do We Need a Blockchain?

Alternative: W3C DIDs & Verifiable Credentials

Decentralized PKI Anchoring instead of Direct Supply Chain Ledger Storage



- **W3C Standards: DIDs & Verifiable Credentials (VCs)**
 - A secure, standard, federated approach that requires **no** global transaction consensus ledger.
- **How it works:**
 - **Issuer:** Manufacturer signs the passport payload (JSON-LD) using

Discussion Prompts for the Audience

- **1. The Liability of the GIGO Link:**

- *Scenario:* A recycler relies on an immutable battery passport stating a battery has 80% cobalt. In reality, it was swapped with a counterfeit lead-acid tag and explodes. Who is legally liable: the manufacturer, the ledger validators, or the recycler?

- **2. The Threat of Consortium Metadata:**

- *Scenario:* Five major car manufacturers operate a permissioned consortium blockchain for automotive passports. Can they analyze the metadata (transaction frequencies, shipping latency) of independent suppliers to force price reductions or squeeze out smaller competitors?

- **3. The Carbon Paradox:**

- *Scenario:* A digital passport tracks a circular t-shirt. Recording raw materials, dyeing, sewing, shipping, and second-hand sales takes 5 blockchain transactions. Does the

Paradigm Comparison

Criteria	Centralized Database (EU Registry)	Consortium Blockchain	Public Blockchain (L2)	DIDs + VCs (Decentralized PKI)
Trust Model	Trust in government	Trust in industry cartel	Math + Miner/Validator set	Cryptographic peer-to-peer
Data Privacy	High (controlled access)	Medium (leaks to rivals)	Extremely low (public)	Excellent (off-chain)
Write Cost	Negligible	Low	High & Unstable	Zero
Scalability	High	Medium	Low	Infinite
GIGO Security	Checked at input	Checked by consortium	Unchecked	Checked by peer-to-peer audit

Thank You!

Let's Open the Critical Discussion

- Is the blockchain hype in circular economy projects just a distraction from standardizing data formats?
- How can we solve the physical-digital linkage without expensive tamper-proof hardware?
- Can circular regulations survive without a centralized state enforcement agency?

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